Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/01/2016
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	09/01/2010	**End Date:**	11/15/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 11/15/2015 per NSSC information (Ed., 1/13/16) NOTE: End date changed to 9/30/2015 per NSSC information (Ed., 9/20/13) NOTE: Extended to 9/30/13 per PI, who also said it would be further extended (Ed., 8/5/13)		
Key Personnel Changes/Previous PI:	Robert Hyers is U.S. Team Member on ESA-sponsored projects, "Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC)," "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)," and "Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL)." Dr. Jonghyun Lee joined UMass Amherst full-time in June 2013. He had previously been a Tufts employee working on our collaboration on LODESTARS and PARSEC.		
COI Name (Institution):			
Grant/Contract No.:	NNX10AR71G		
Performance Goal No.:			

Performance Goal Text:

Task Description:

Funding is for Dr. Hyers's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted.
Main project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany

Subprojects:

1. Transformation Kinetics-ISS (International Space Station). Task description: Using results from thermophysical property evaluation and combining these with the Magnetohydrodynamic (MHD) convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection.

2. Thermophysical Properties. Task description: The density, viscosity, and surface tension of the flight samples will be measured in flight for a range of temperatures. Density will be determined by the video metrology, and surface tension and viscosity by the oscillating drop method. This WP includes determination of parameter modifications required for these measurements as well as collection, reduction, and analysis of the resulting data. The resulting data will support the whole range of theoretical and modeling efforts in this proposal.

3. Modeling of Macroconvection. Task description: Magnetohydrodynamic (MHD) model of macroconvection in the samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theorerical and modeling efforts. Collaboration on development of parameters for other work packages to enable the use of convective state as an experimental parameter.

Main project: Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany.

Subproject:

Modeling of macroconvection: Magnetohydrodynamic modelling of macroconvection in Ti-Zr-Ni samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theorerical and modeling efforts.

Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany

Subprojects:

Long Duration Microgravity Experiments - ISS. Evaluation of microgravity experiments for fluid flow investigations.

MODELLING A: Thermodynamics, Phase Selection. Modelling of nucleation under various flow conditions for steel alloys.

MODELLING B: Non-contact thermophysical. Modeling of magnetohydrodynamic convection and shear in 1-g, microgravity for 4-6 alloys. Modeling of Marangoni convection in 1-g ESL drops for comparison with microgravity experiments. Modeling of temperature distribution in 1-g and microgravity samples, 4-6 alloys. Input into design of experimental parameters for 4-6 Alloys.

ED. NOTE (7/18/2016): Continued by "Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC (NNX16AB40G)" with the same Principal Investigator, Dr. Robert Hyers. See that project for continued reporting.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This project consists of modeling in support of three different flight experiments:
- PARSEC - Peritectic Alloy Rapid Solidification with EM Convection

- ThermoLAB - Thermophysical Properties of Liquid Metallic Alloys

- ICOPROSOL - Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order

In PARSEC, we are investigating the effect of fluid flow on the solidification path of certain alloys. Control of the solidification path would enable tailoring of the microstructure and properties of metal parts for specific applications.

In ThermoLAB, we are investigating the thermophysical properties of high-temperature materials. Many of these materials are used commercially. A better understanding of the physical properties will allow more efficient and more reliable production of metallic parts using these alloys. Other alloys are chosen because of scientific interest in their properties, which may illuminate fundamental physics related to the formation of bulk metallic glasses and quasicrystals.

In ICOPROSOL, we are investigating the nucleation and growth of quasicrystals, and the effect of atomic-scale order on the macroscopic properties of these alloys. This fundamental investigation may improve our ability to tailor the microstructure of metals for commercial applications.

Task Progress:

Highlights of Work Accomplished During Grant Period 2010-15.
MHD Modeling

• Validated magnetic model with new MSL-EML flight coils against experiments by Dr. Georg Lohoefer, DLR, Cologne in 2011.

• Rebuilt and validated magnetohydrodynamic model for laminar internal flow in electromagnetically levitated droplets in COMSOL Multiphysics in 2011.

• Rebuilt and validated Marangoni convection model with simple boundary conditions in COMSOL Multiphysics in 2011.

• First experimental validation of magnitude and direction of turbulent magnetohydrodynamic flows in EML; paper submitted to Met. Trans.

• Trained personnel at Tufts University in MHD modeling methodology and techniques for the efficient support of the space experiments.

• Predicted an accessible range of convection in the reference Zr sample during the MSL-EML experiment.

• Ms. Jie Zhao investigated the turbulent transition in an electromagnetically-levitated NiAl3 droplet for her thesis project in collaboration with DLR.

• Supported the magnetohydrodynamic simulations for the prediction of phase selection during solidification for PARSEC.

Ground support for MSL-EML experiments

• Precision-ground samples for MSL-EML Batch 1. Delivered samples to U. Ulm, who delivered them to Astrium for integration with the flight facility.

• Supported the measurements of thermophysical properties of Fe-Co systems including the mass evaporation calculation for PARSEC.

• Theoretically calculated the amount of mass evaporation of the FeCrNi sample tested during FCE1.2 to support the subsequent Batch1 processing of ThermoLAB and PARSEC samples.

• Supported the measurements of thermophysical properties of Fe-Co systems including the mass evaporation calculation for PARSEC.

• Ms. Jie Zhao is investigating the influence of oxygen on the thermophysical properties of molten Zirconium for her Ph.D. thesis project in collaboration with NASA MSFC.

Supporting group researchers: • Hired Ms. Briana Tomboulian MS/PhD student at UMass to conduct flow modeling tasks in 2010. Ms. Tomboulian was selected for a NASA Science and Technology Research Fellowship, which will support her PhD Research. • Hired Ms. Jie Zhao, a MS/PhD student, to work on improving our prediction of the turbulent transition in EML flows. She is doing modeling and analysis of prior experiments in EML in parabolic and sounding rocket flights. • Ms. Christina Rizer won a summer undergraduate fellowship from the Massachusetts Space Grant Consortium to work with Ms. Zhao on turbulent transition. • Hired Dr. Jonghyun Lee full-time at UMass Amherst. Dr. Lee had been working at Tufts, supporting our collaboration on LODESTARS and PARSEC.

Meetings and conferences

• Attended ten (3rd to 12th) MSL-EML Working Group Meetings (2010-2015).

• Invited member ESA International Topical Team on Thermophysical Properties (ThermoLab-ISS kick-off meeting) at ESA/ESTEC 11/21/2010.

• Lead organizer for symposium “Materials Research in Microgravity”, TMS (The Minerals, Metals & Materials Society) Annual Meeting 2012, Orlando, FL.

• Gave an invited presentation on the subject research at a joint seminar between Mechanical Engineering and Physics at Washington University, St. Louis, MO, November 7, 2012. On-site meetings with Prof. K.F. Kelton about ICOPROSOL.

• Invited speaker, Workshop on Dynamics, Properties, and Solidification of Liquid Alloys, German Aerospace Center, Cologne, Germany, July 7, 2011.

• Presentation at the International Symposium on Physical Sciences in Space, Bad Godesburg, Germany, July 2011.

• Invited presentation, 50th AIAA Aerospace Sciences Meeting, Nashville, TN, January 2012. Meetings at NASA MSFC with PARSEC team.

• Gave an invited presentation on the subject research at the Physics Colloquium, Amherst College, March, 2012.

• Attended PI visit to DLR/MUSC for development of the Science Protocol, May 2012.

• Meetings in Ulm, Germany, with ThermoLab team, May 2012.

• Attended 6th MSL-EML Working Group at DLR, June 2012, and associated PI meetings.

• Lead organizer for symposium “Materials Research in Microgravity”, TMS Annual Meeting 2012, Orlando, FL.

• Co- organizer for symposium “Materials Research in Reduced Gravity”, TMS Annual Meeting 2013, San Antonio, TX.

• Participated in thermophysical property measurements at NASA MSFC ESL, August 2012.

• Co- organizer for symposium “Materials Research in Reduced Gravity”, TMS Annual Meeting 2013, San Antonio, TX.

	• Attended MSL-EML Investigators Working Group meeting, Friedrichshafen, Germany, November 2012. • Attended MSL-EML Investigators Working Group meeting, San Antonio, TX, March, 2013. • Represented UMass at the Massachusetts Space Grant Consortium Annual Meeting, Boston, MA, April 2013. • Attended the Herlach symposium and MSL-EML Investigators Working Group meeting, Cologne, Germany, September 2014. • Attended 12th MSL-EML Investigators Working Group meeting, Cologne, Germany, May 2015. Students Supported: Christina Rizer, undergraduate research 2013. Briana Tomboulian, PhD, 2014. Xiao Ye, PhD, 2014. Jie Zhao, MS 2014, PhD expected 2016. ED. NOTE (7/18/2016): Continued by "Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC (NNX16AB40G)" with the same Principal Investigator, Dr. Robert Hyers. See that project for continued reporting.
Bibliography Type:	Description: (Last Updated: 03/30/2026)
Articles in Peer-reviewed Journals	Lee J, Rodriguez JE, Hyers RW, Matson DM. "Measurement of density of Fe-Co alloys using electrostatic levitation." Metallurgical and Materials Transactions B. 2015 Dec;46B:2470-5. http://dx.doi.org/10.1007/s11663-015-0434-7 , Dec-2015
Articles in Peer-reviewed Journals	Lee J, Xiao X, Matson DM, Hyers RW. "Numerical prediction of the accessible convection range for an electromagnetically levitated Fe50Co50 droplet in space." Metallurgical and Materials Transactions B. 2015 Feb;46(1):199-207. First online: 17 September 2014. http://dx.doi.org/10.1007/s11663-014-0178-9 , Feb-2015